Bahram Afsharmanesh

Curriculum Vitae

University of Georgia
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Research Interests

High-performance computing and machine learning systems; heterogeneous GPU scheduling and distributed training; GPU kernels and collective communication; communication–computation overlap, quantization, and performance modeling for large language model inference.

Education

University of Georgia 2023–Present
Ph.D. in Computer Science, in progress GPA: 3.95/4.00
Advisor: Prof. Gagan Agrawal.

Sharif University of Technology 2012–2014
M.Sc. in Mechanical Engineering GPA: 3.81/4.00

Iran University of Science and Technology 2008–2012
B.Sc. in Mechanical Engineering GPA: 3.76/4.00

Selected Publications

[1] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **LiGO: Low-Bit GEMM–AllReduce Overlap for Tensor-Parallel LLM Inference.** *IEEE International Parallel & Distributed Processing Symposium (IPDPS'27)*. Submitted.

[2] AmirAli Mirian, **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, Wei Niu, and Gagan Agrawal. **Fast Sparse Linear Algebra for LLMs with Contextual Sparsity.** *IEEE International Parallel & Distributed Processing Symposium (IPDPS'27)*. Submitted.

[3] **Bahram Afsharmanesh**, Md Musfiqur Rahman Sanim, AmirAli Mirian, and Gagan Agrawal. **HeteroSched: Co-Optimizing Scheduling and Parallelization for Deep Learning Workloads for Heterogeneous GPU Clusters.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026.

[4] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Wei Niu, and Gagan Agrawal. **Mask-Aware Execution for Efficient JEPA Training.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026.

[5] Md Musfiqur Rahman Sanim, Zhihao Shu, **Bahram Afsharmanesh**, AmirAli Mirian, Jiexiong Guan, Wei Niu, Bin Ren, and Gagan Agrawal. **Optimizing 3D Gaussian Splattering for Mobile GPUs.** *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2025. [Preprint]

Professional Service

Artifact Evaluation Committee Member 2027
Symposium on Principles and Practice of Parallel Programming (PPoPP).
Artifact Evaluation Committee Member 2026
International Conference on Parallel Architectures and Compilation Techniques (PACT).

Research Experience

Graduate Research Assistant Aug. 2023–Present
University of Georgia; advisor: Prof. Gagan Agrawal

- Designed HeteroSched to jointly optimize scheduling and parallelization for heterogeneous GPU clusters; evaluated with performance simulation and physical experiments.
- Developed CUDA/C++ and MSCCL++ collective-communication kernels, fine-grained communication–computation overlap, and quantized AllReduce for distributed LLM inference over PCIe and NVLink.
- Built microbenchmarks and analytical performance models; used Nsight Systems and Nsight Compute to study synchronization, memory bandwidth, kernel occupancy, and end-to-end performance.

Research Assistant May 2020–June 2021
Amirkabir University of Technology

- Investigated machine learning as a function approximator for numerical PDE solvers; developed MATLAB applications for mechanical-system analysis under static, dynamic, and thermal loads.

Research Assistant Sept. 2012–Sept. 2015
Sharif University of Technology

- Developed PDE solvers in MATLAB and molecular dynamics algorithms in Fortran; used genetic algorithms to calibrate numerical models against molecular dynamics simulations.

Undergraduate Researcher Sept. 2010–Sept. 2012
Iran University of Science and Technology; advisor: Prof. S. M. Hasheminejad

- Solved PDEs for laminated plates using MATLAB and Mathematica.

Industry Experience

Data Scientist Aug. 2021–Dec. 2022
Baroline

- Built spam-detection and named-entity-recognition models with PyTorch, BERT, and scikit-learn; deployed inference APIs using FastAPI, Docker, and Kafka, reducing response latency by 25%.
- Developed data pipelines and warehouses with Airflow and PostgreSQL, and analytics dashboards with Pandas, Streamlit, Power BI, and Metabase.

R&D Engineer Jan. 2016–Apr. 2020
MAPNA PARS Company

- Built MATLAB toolboxes for system identification and eigenvalue analysis; developed regression models to predict component behavior and accelerate engineering design workflows.

Selected Technical Projects

CUDA Parallel Prefix Scan. Implemented a hierarchical GPU scan using shared memory, custom kernels, multilevel residue computation, and uniform addition for large inputs.

Deep Neural Network Pruning. Implemented one-shot and iterative magnitude pruning, including unstructured and filter pruning; evaluated sparsity and model-performance tradeoffs.

Technical Skills

Programming	C, C++, Python, CUDA, Fortran, SQL, MATLAB, Mathematica, Bash
GPU and parallel	MSCCL++, NCCL, CUTLASS/CuTe, Triton, MPI, OpenMP
ML and data	PyTorch, JAX, scikit-learn, NumPy, Pandas, FastAPI, Kafka, Airflow
Performance	Nsight Systems, Nsight Compute, profiling, benchmarking, analytical performance modeling, roofline analysis
Development	Linux, Git, Docker, Make/CMake, CI/CD, AWS, LaTeX